Cheating sheet

$$\mathrm{Der}(\mathcal{O}_X, M) \cong \mathrm{Hom}(\Omega_X, M)$$

$$\mathrm{Def}_C(\mathbb{C}[\varepsilon]/\varepsilon^2) \cong Ext^1(\Omega_C, \mathcal{O}_C) \cong H^0(\omega_C^{\otimes 2}) \cong \mathbb{C}^{3g-3}$$

infinitesimal automorphisms $\leftrightarrow Ext^0(\Omega_X, \mathcal{O}_X) \cong H^0(T_X)$

infinitesimal deformations $\leftrightarrow Ext^1(\Omega_X, \mathcal{O}_X) \cong H^1(T_X)$

obstructions $\leftrightarrow Ext^2(\Omega_X, \mathcal{O}_X) \cong H^2(T_X)$

$\Omega_X \rightsquigarrow L_X$ cotangent complex

clutching/gluing: $\overline{\mathcal{M}}_{g_1,n_1+1} \times \overline{\mathcal{M}}_{g_2,n_2+1} \longrightarrow \overline{\mathcal{M}}_{g_1+g_2,n_1+n_2}$



forget: $\pi: \overline{\mathcal{M}}_{g,n+1} \longrightarrow \overline{\mathcal{M}}_{g,n}$

$\overline{\mathcal{C}}_{g,n} \xrightarrow{\cong} \overline{\mathcal{M}}_{g,n+1}$, $\tilde{\pi}$, π, $\overline{\mathcal{M}}_{g,n}$

$\psi_i = c_1(\mathcal{L}_i) \in H^2(\overline{\mathcal{M}}_{g,n})$ $\quad (\mathcal{L}_i)_{[C,p_1,\dots,p_n]} = T_{p_i}^* C$

$\int_{\overline{\mathcal{M}}_{g,n}} \psi_1^{a_1} \cdots \psi_n^{a_n}$ $(\sum a_i = 3g-3+n)$: Witten Kontsevich intersection numbers

$\lambda_i = c_i(\mathbb{E}) \in H^{2i}(\overline{\mathcal{M}}_{g,n})$ $\quad (\mathbb{E})_{[C,p_1,\dots,p_n]} = H^0(C, \omega_C)$

$\mathbb{E} = \tilde{\pi}_* \omega_\pi$

$\kappa_a = \pi_*(\psi_{n+1}^{a+1}) \in H^{2a}(\overline{\mathcal{M}}_{g,n})$ $\quad \pi: \overline{\mathcal{M}}_{g,n+1} \to \overline{\mathcal{M}}_{g,n}$

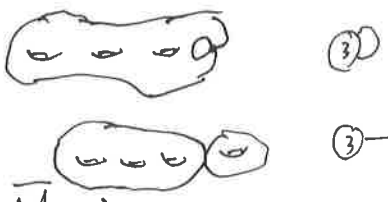
boundary strata:

$\Delta = \overline{\mathcal{M}}_{g,n} \setminus \mathcal{M}_{g,n} = \Delta_{1\dots} \cup \dots$ (codim 1 $\cup$ codim >1 $\cup$ codim >1 $\cup \dots$)

$\delta_* = [\Delta_*] \in H^2(\overline{\mathcal{M}}_{g,n})$

$\Delta_{irr} = \mathrm{Im}\left(\overline{\mathcal{M}}_{g-1,2} \to \overline{\mathcal{M}}_g\right)$

$\Delta_i = \mathrm{Im}\left(\overline{\mathcal{M}}_{i,1} \times \overline{\mathcal{M}}_{g-i,1} \to \overline{\mathcal{M}}_g\right)$



$\Delta_{i,S} = \mathrm{Im}\left(\overline{\mathcal{M}}_{i,S\cup\{*\}} \times \overline{\mathcal{M}}_{g-i,S^c\cup\{*\}} \to \overline{\mathcal{M}}_{g,n}\right)$ $\quad S \subseteq \{1,\dots,n\}$

$$\Delta = \bigcup_\Gamma \Delta_\Gamma = \Delta_{irr} \cup \bigcup_{i,S} \Delta_{i,S} \cup \text{codim} > 1 \dots$$

$R^*(\overline{\mathcal{M}}_{g,n}) \subset A^*(\overline{\mathcal{M}}_{g,n})$

$$\begin{array}{ccc} [\operatorname{Spec} A/G] & & \mathcal{X} \\ \downarrow & & \downarrow \phi \\ \operatorname{Spec} A^G & & X \end{array}$$

$$\phi_* \mathcal{O}_{\mathcal{X}} = \mathcal{O}_X$$

+ cohomologically affine

good moduli space = stack-theoretic generalization of GIT stack

E.g. $X = \{xy = 0\} \subseteq \mathbb{A}^2$

$$L_X \cong \left[\mathcal{O}_X \longrightarrow \mathcal{O}_X\,dx \oplus \mathcal{O}_X\,dy \right]$$

$$\begin{array}{ccc} -1 & \longmapsto & y\,dx + x\,dy \\ 1 & & \\ \text{A} & & \text{B} \end{array}$$

$$RHom(L_X, \mathcal{O}_X) \cong \left[\longrightarrow \mathcal{O}_X\,\partial_x \oplus \mathcal{O}_X\,\partial_y \longrightarrow \mathcal{O}_X \right]$$

$$\begin{array}{ccc} 0 & & 1 \\ \partial_x & \longmapsto & y \\ \partial_y & \longmapsto & x \\ (a,b) & \longmapsto & ay + bx \end{array}$$

$$\begin{array}{ccc} \{ & & \{ \\ Ext^\circ(L_X, \mathcal{O}_X) & \xrightarrow{(xf, -yf)} & Ext^1(L_X, \mathcal{O}_X) \\ \text{sits} & & \| \\ \mathcal{O}_X & \xrightarrow{f} & \mathcal{O}_X/(x,y) = k \end{array}$$

For $[C, p_1, \dots, p_n] \in \overline{M}_{g,n}$,

finite auto $\iff$ $\omega_C(\sum p_i)$ is ample

$K_X + \Delta$

MMP = systematic birational simplification controlled by $K_X+\Delta$

$K_X+\Delta$ nef $\Rightarrow$ minimal model

$K_X+\Delta$ nef + big $\overset{\text{often}}{\Rightarrow}$ canonical model

$$X_{can} = \mathrm{Proj}\left(\bigoplus_{m\geq 0} H^0(m(K_X+\Delta))\right)$$

where $K_X+\Delta$ is ample

discrepancies, 差異 $f: Y \to X$ resolution.

$$K_Y + \Delta_Y = f^*(K_X+\Delta_X) + \sum_{E\in \mathrm{Div}(Y)} a(E,X,\Delta_X)E$$

$a(E)$

smooth $\subset$ klt $\subset$ lc

Kawamata log terminal

log canonical

$a(E)=0$ $\quad a(E)>-1$ $\quad a(E)\geq -1$

$\nu: X^\nu \to X$ morally

X is slc $\iff$ (X^ν, D) is lc

semi-log canonical

slc: allow nodal reducible curves.

$(C, \Sigma p_i)$ has nodal sing $\omega_C(\Sigma p_i)$ ample $\overline{M}_{g,n}$

(X, Δ) is slc $K_X + \Delta$ is ample KSBA moduli

KSBA: Kollár - Shepherd - Barron - Alexeev

sm curves $\subset$ stable

sm can pol varieties $\subset$ slc varieties with K_X ample

KSBA stability: (X, Δ) is in KSBA moduli?

K-stability: (X, L) behave under 1-parameter degenerations?

e.g. $L = -K_X$, X Fano

$$\begin{array}{ccc} \mathcal{X} & \curvearrowright \mathbb{G}_m & X \\ \downarrow & \curvearrowright \mathbb{G}_m & \downarrow \\ \mathbb{A}^1 & & 1 \end{array} \Rightarrow DF(\mathcal{X}) : \text{Donaldson - Futaki invariant}$$

test configuration

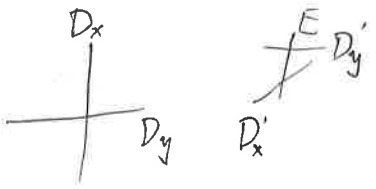
$x \in [X/G]$ semistable $\Leftrightarrow \lim_{t \to 0} \lambda(t) \cdot x \neq 0$ for all $\lambda \in PS(G)$, $\mu(x, \lambda) \geq 0$

X: K-semistable $\Leftrightarrow DF(\mathcal{X}) \geq 0$ for all test configurations

X: K-polystable $\Leftrightarrow \exists$ canonical metrics

e.g. Kähler - Einstein metric when X is Fano

E.g. $X = \mathbb{A}^2$, $\Delta = D_x + D_y$

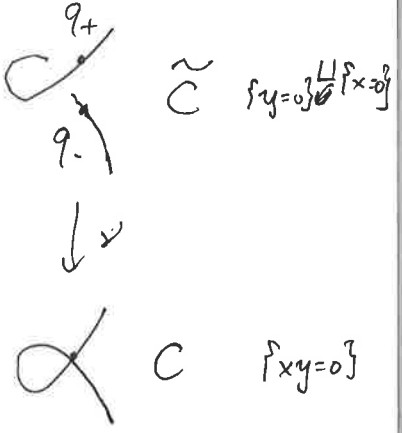


$$f^*K_X = K_Y - E$$

$$f^*(D_x + D_y) = D_x' + D_y' + 2E$$

$$K_Y + D_x' + D_y' = f^*(K_X + D_x + D_y) - E$$

E.g. $\nu^* \omega_C \simeq \omega_{\tilde{C}}(q_+ + q_-)$



$$K_{\tilde{C}} + \underbrace{q_+ + q_-}_{D} = \nu^* K_C$$

$$a(q_+, \tilde{C}, D) = a(q_-, \tilde{C}, D) = -1$$

coefficient ≤ 1.

$\Rightarrow$ node C is slc

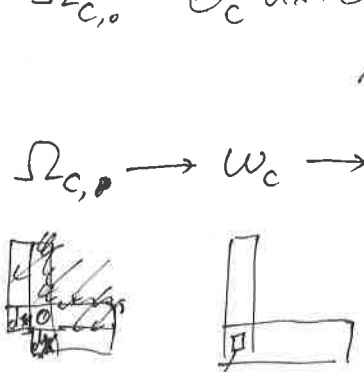
$$\omega_{C,0} = \mathcal{O}_C \cdot \left(\frac{dx}{x}, -\frac{dy}{y}\right)$$

$$\Omega_{C,0} = \mathcal{O}_C \cdot dx + \mathcal{O}_C \cdot dy \,/\, (x\,dy + y\,dx)$$

$$0 \to \tau \to \Omega_{C,0} \to \omega_C \to \mathbb{C}\left(\frac{dx}{x} - \frac{dy}{y}\right) \to 0$$

$\mathbb{C} \cdot x\,dy$

0



$\square$

$\frac{dx}{x} - \frac{dy}{y}$

q

$\{y^2 = x^3\}$ $\quad 2y\,dy = 3x^2\,dx$

$$\omega_{C,q} = \mathcal{O}_C \cdot \left(\frac{dx}{2y}, -\frac{dy}{3x^2}\right)$$

$$\begin{cases} x = t^2 \\ y = t^3 \end{cases} \rightsquigarrow \frac{dt}{t^2}$$

$\nu^* \omega_C = \omega_{\tilde{C}}(2q)$

$a(q, \tilde{C}, D) = 2$.

A: $f, g \in \mathbb{C}[[t^2, t^3]]$

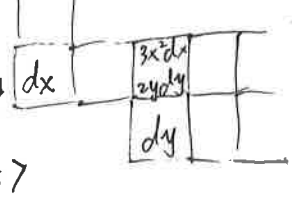
$f\,dx + g\,dy = 0$

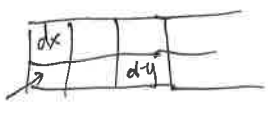
$\begin{cases} x = t^2 \\ y = t^3 \end{cases} \Rightarrow 2f + 3tg = 0$

$$0 \to \tau \to \Omega_{C,q} \to \omega_{\tilde{C}} \to \mathbb{C}^2 \to 0$$

τ: $\langle 2x\,dy - 3y\,dx,\ 2x^2\,dy - 3xy\,dx \rangle_{\mathbb{C}}$

$\omega_{\tilde{C}}$: $\left\langle \frac{dx}{2y}, \frac{x\,dx}{2y} \right\rangle_{\mathbb{C}}$





$\frac{dx}{2y}$ $\frac{dy}{3x^2}$

$S = S_g$ $\quad (X, f)$ $\quad f: S \to X$ topo iso

$\mathcal{T}_g = \{$cplx structures on $S\}/_{\text{isotopy}}$ marked curves

$\mathcal{M}_g = \mathcal{T}_g / Mod_g$

$Mod_g = Diff^+(S)/Diff_0(S)$ discrete gp

↑ orientation preserving ↑ conn components

$X \cong \mathcal{H}/\Gamma$

$$\begin{array}{ccc} \tilde{S} & \longrightarrow & \mathcal{H} \\ \downarrow & & \downarrow \\ S & \xrightarrow{f} & X \end{array}$$

$\rho: \pi_1(S) \to \pi_1(X) \cong \Gamma \subseteq PSL_2(\mathbb{R})$

$\mathcal{X}_G(S) = Hom(\pi_1(S), G)/\!/G$ depends on S, no cplx structure

Fact: $\mathcal{T}_g$ is a connected component of $\mathcal{X}_{PSL_2(\mathbb{R})}(S)$

$\mathcal{X}_G(S) = Rep_G(\pi_1(S)) \longleftrightarrow LS_G(S) \longleftrightarrow$ flat G-bundle on S

harmonic metric $\longleftrightarrow$ polystable G-Higgs bundle

not cplx structure $\mathcal{M}_{Higgs}(X; G)$

E.g. $G = GL_n(\mathbb{C})$,

(E, ϕ) is a Higgs bundle, if

- E: holo v.b. on X
- $\phi \in H^0(X; End(E) \otimes K_X)$

As real analytic spaces or hyperkähler mflds

Fix D flat connection on E,

choose a harmonic metric, then

$$D = \bar{\partial}_E + \partial_E + \phi + \phi^\dagger$$

$\bar{\partial}_E$: holo bd structure on E; ϕ: Higgs field.

Hitchin fibration:

$$\phi : E \longrightarrow E \otimes K_X$$

$$h(E,\phi) = \det(\lambda \mathrm{Id} - \phi) = \lambda^n + a_1 \lambda^{n-1} + \cdots + a_n \qquad a_i \in H^0(K_X^i)$$

$$h : \mathcal{M}_{\mathrm{Higgs}} \longrightarrow \bigoplus_{i=1}^{n} H^0(K_X^i)$$

Hitchin base

$a_1 = 0$ if SL_n case

total space. L

K_X $\quad \mathrm{Tot}(K_X) \supset \Sigma \qquad \lambda \in H^0(\mathrm{Tot}(K_X), \pi^* K_X)$

X $\quad \pi$ (n:1) $\qquad \Sigma = \{(x,\xi) \in \mathrm{Tot}(K_X) \mid \det(\lambda - \phi_x) = 0\}$

spectral curve

Beauvill-Narasimhan-Ramanan Spectral correspondence.

generic Hitchin fiber: $\mathrm{Jac}(\Sigma)$ or a Prym variety.

rk 1 torsion-free sheaf $\quad SL_n$: $\ker'(\mathrm{Jac}(\Sigma) \to \mathrm{Jac}(C))$

l.b. on Σ: $\quad L_{(x,\lambda)} = \ker(\phi_x - \lambda \mathrm{id}) \subseteq E_x$

$$(\Sigma, L) \rightsquigarrow E := \pi_* L$$

λ,

$$\times\lambda : L \xrightarrow{\exists\, \lambda \in H^0(\Sigma, \pi^* K_X)\ \text{canonical}} L \otimes \pi^* K_X$$

$$\phi : \underbrace{\pi_* L}_{E} \longrightarrow \underbrace{\pi_* L \otimes K_X}_{E}$$

$H^0(X; K_X^2)$ - Hitchin base for SL_2

- $T^*_X \mathcal{T}_g = T^*_X \mathcal{M}_g$

$H^0(K^2)$ parameterizes a distinguished family of Higgs bds.

E.g. $G = SL_2$, $E = K^{\frac{1}{2}} \oplus K^{-\frac{1}{2}}$, ϕ

$\phi = \begin{pmatrix} 0 & q \\ 1 & 0 \end{pmatrix}$ $\quad q \in H^0(K^2)$

$\rightsquigarrow$ Hitchin component. ep. $\mathcal{T}_g$ if $G = PSL_2(\mathbb{R})$